

Functional Annotation of PP_1907 (UniProt Q88LM3) — *Pseudomonas putida* KT2440

Gene: PP_1907 (OrderedLocusName) · **UniProt:** Q88LM3 · **RefSeq:** WP_010952931.1 · **Protein length:** 224 aa **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / NCIMB 11950 / KT2440) · **Genome position:** NC_002947.4, 2,152,277–2,152,951 (+ strand)

1. Summary (Answer to the Research Question)

PP_1907 encodes a **small, soluble, cytoplasmic enzyme of the haloacid dehalogenase (HAD) hydrolase superfamily, subfamily IA** — a **Mg²⁺-dependent phosphomonoesterase (phosphatase)**. Its aspartate-nucleophile catalytic machinery is fully intact, so it is confidently a functional phosphatase that hydrolyses a phosphomonoester substrate via a phospho-aspartyl enzyme intermediate. Every automated functional-classification resource (KEGG ortholog **K01091**, PANTHER PTHR43434, UniProt/TreeGraft) assigns it the specific activity **2-phosphoglycolate phosphatase (Gph; EC 3.1.3.18)**, catalysing:



placing it in **glyoxylate and dicarboxylate metabolism** (KEGG pathway ppu00630). By analogy to the biochemically characterised *Escherichia coli* Gph, the physiological role is a **cytoplasmic housekeeping "salvage" reaction** that disposes of 2-phosphoglycolate — most notably the 2-phosphoglycolate liberated during repair of oxidatively damaged 3'-phosphoglycolate DNA ends — recovering the two-carbon unit as glycolate for central metabolism.

Important caveat: This assignment is **inferential (bioinformatic / structural / evolutionary), not experimental**. PP_1907 itself has never been biochemically characterised, and it is **one of three paralogous HAD-IA phosphatases** in *P. putida* that carry the K01091 annotation. A closer paralog (**PP_0416**, 43% identical to *E. coli* Gph) is the more likely *principal*

2-phosphoglycolate phosphatase, so PP_1907's precise physiological substrate should be regarded as a well-motivated prediction that may include 2-phosphoglycolate and/or related small phosphometabolites, potentially with partial functional redundancy.

2. Identity Verification

The requested identifiers are internally consistent and unambiguous: - Gene symbol **PP_1907** ↔ protein description "Hydrolase, haloacid dehalogenase-like family" ↔ organism *P. putida* KT2440 — all match across UniProt (Q88LM3), NCBI (AAN67525.1 / WP_010952931.1), KEGG (ppu:PP_1907) and STRING (160488.PP_1907). - The listed InterPro domains (HAD-SF_hydro_IA IPR006439; HAD_2 IPR041492; HAD-like_sf IPR036412; HAD_sf IPR023214) exactly match what is found for this accession. - No conflicting literature for a different gene of the same symbol exists (PP_ locus tags are unique to the *P. putida* KT2440 genome).

There is **no ambiguity of identity**; the only uncertainty concerns the *precise substrate*, discussed below.

3. Molecular Function — What Reaction Is Catalysed, and Substrate Specificity

3.1 Enzyme superfamily and catalytic mechanism

PP_1907 belongs to the **HAD superfamily** (eggNOG **COG0546**; Pfam **PF13419** HAD_2; SUPFAM SSF56784; NCBIfam TIGR01509/TIGR01549 "HAD-SF-IA"). HAD-superfamily enzymes are overwhelmingly **phosphotransferases/phosphohydrolases** that use an **aspartate nucleophile** to form a covalent phospho-aspartyl intermediate, with an essential **Mg²⁺** cofactor. All four diagnostic HAD sequence motifs are present in PP_1907:

Motif	Sequence in PP_1907	Residue(s)	Role
I (DxDxT/V)	¹⁰ F-D11-W-D13-G-T15-L	Asp11 (nucleophile), Asp13 (general acid/base), Thr15	Nucleophilic attack; proton shuttle; phosphate binding
II (S/T)	cap/core loop	conserved Ser/Thr	H-bonds transferring phosphoryl group
III (K)	¹¹⁰ LAVA-T-G-K113	Lys113	Neutralises phosphate negative charge
IV (GDxxxD / DD)	¹⁶⁷ G-D168-S-A-F-D172	Asp168, Asp172	Coordinate catalytic Mg ²⁺

UniProt lists **Mg²⁺ (ChEBI:18420)** as the cofactor (ARBA rule), consistent with this mechanism. This constitutes strong structural/bioinformatic evidence that PP_1907 is a *bona fide* Mg²⁺-dependent phosphatase, independent of any substrate assignment.

Residue-level conservation with a characterized enzyme. A pairwise global alignment of PP_1907 against the biochemically characterised *E. coli* Gph (P32662) shows that **all six catalytic residues align one-to-one to identical residues** (Asp11, Asp13, Thr15, Lys113, Asp168, Asp172 — every position invariant), even though overall identity is only ~29%. Because 2-phosphoglycolate phosphatases of diverse origins share highly conserved catalytic features implying a common mechanism (Pellicer et al., 2003,),

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this invariance is strong evolutionary evidence that PP_1907 is a **catalytically competent phosphatase using the same aspartyl-phosphotransferase mechanism.**

3.2 Structural architecture and substrate-size class

The protein has the canonical **two-domain HAD-IA fold**: a Rossmann-like α/β core (**Gene3D 3.40.50.1000**) plus a small α -helical "**C1 cap**" inserted immediately after Motif I (**Gene3D 1.10.150.240** "putative phosphatase domain 2"; **IPR023198** "phosphoglycolate-phosphatase-like domain 2"; cap region $\approx$ residues 17–95). In the HAD superfamily the **cap domain is the principal determinant of substrate specificity and catalytic efficiency** (Daughtry et al., 2013,

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Biswas et al., 2009, [P 19726684](#)).

A **small C1 cap of this type is characteristic of subfamily-IA phosphatases that act on small, mono-anionic phosphometabolites** — precisely the class that includes 2-phosphoglycolate phosphatase.

3.3 Specific substrate assignment and its confidence

Multiple independent resources converge on **2-phosphoglycolate phosphatase (EC 3.1.3.18)**: - **KEGG**: ppu:PP_1907 → **K01091** (phosphoglycolate phosphatase), pathway ppu00630 (glyoxylate/dicarboxylate metabolism). - **PANTHER**: family **PTHR43434** = **PHOSPHOGLYCOLATE PHOSPHATASE**. - **UniProt/TreeGrafter** **GO**: molecular function **GO:0008967 phosphoglycolate phosphatase activity**. - **Sequence homology**: among characterised HAD-IA enzymes, PP_1907 is **closest to the experimentally validated *E. coli* Gph** (2-phosphoglycolate phosphatase, P32662; 29.0% global identity) — above *E. coli* YniC (23.6%), YidA sugar-phosphatase (20.4%), NagD (17.6%), and human PGP (24.6%). The *E. coli* Gph enzyme is **highly specific for 2-phosphoglycolate with high catalytic efficiency** (Pellicer et al., 2003, [P 13129953](#)).

Tempering evidence (see §5): *P. putida* KT2440 contains **three** K01091-annotated paralogs (PP_0094, PP_0416, PP_1907). **PP_0416** is 43.1% identical to *E. coli* Gph and is explicitly named "Phosphoglycolate phosphatase", making it the strongest candidate for the organism's principal 2-phosphoglycolate phosphatase; PP_0094 (25% identity) is annotated "5'-nucleotidase". The three paralogs share only 25–28% pairwise identity (ancient divergence). Because HAD-IA phosphatases frequently display **broad, overlapping substrate spectra** (Kuznetsova et al., 2006, [P 16990279](#)),

PP_1907 may act on 2-phosphoglycolate and/or additional small phosphorylated metabolites; the single-substrate label is a prediction, not a measurement.

4. Localization and Biological Process / Pathway

4.1 Subcellular localization

PP_1907 is predicted **cytoplasmic** (GO:0005829 cytosol). It is a 224-residue soluble protein with **no signal peptide and no transmembrane segments**, consistent with the soluble, cytoplasmic nature of HAD-IA small-molecule phosphatases. Its function is therefore carried out **intracellularly, in the cytosol**.

4.2 Pathway context and physiological role

The reaction sits at the entry of **glyoxylate/dicarboxylate metabolism**. 2-Phosphoglycolate is a two-carbon phospho-metabolite; its dephosphorylation yields **glycolate**, which *P. putida* KT2440 can further oxidise and assimilate — the organism encodes a complete downstream route: **glycolate oxidase GlcDEF** (PP_3744/PP_3746/PP_3747; glycolate → glyoxylate), **glyoxylate carboligase Gcl** (PP_4297), and **malate synthase** (PP_0356), feeding carbon into central metabolism.

The **biological rationale** comes from the characterised *E. coli* ortholog: Gph is a **constitutively expressed housekeeping enzyme** that performs the **dissimilation/salvage of intracellular 2-phosphoglycolate formed during DNA repair of 3'-phosphoglycolate ends** — DNA lesions generated by oxidative and radiomimetic damage (e.g., bleomycin) — converting it to glycolate that re-enters general metabolism (Pellicer et al., 2003,).

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This directly explains the **GO:0006281 (DNA repair)** annotation attached to PP_1907: the enzyme is not a DNA-repair protein itself but processes a **by-product** of DNA-end repair. (In photosynthetic/autotrophic organisms 2-phosphoglycolate arises mainly from RuBisCO oxygenase activity; *P. putida* is a heterotroph, so the DNA-repair/oxidative source is the physiologically relevant one.)

4.3 Genomic context

PP_1907's neighbours (PP_1904 *murB*, PP_1905 *rne*/RNase E, PP_1906 *rluC*, PP_1908 protease IV, PP_1909 pyrophosphatase) are **unrelated housekeeping genes**; the gene is **not part of a substrate-specific operon**, consistent with the constitutive housekeeping expression pattern reported for *E. coli gph*. STRING functional-partner networks link PP_1907 to the glycolate-

oxidation machinery (GlcDEF) and glyoxylate reductases (GhrB, HprA), but these associations are **near-identical for all three paralogs** and therefore reflect **family-level co-annotation rather than PP_1907-specific evidence**.

5. Hypotheses: Supported and Refuted / Tempered

Supported 1. *PP_1907 is a Mg^{2+} -dependent HAD-superfamily subfamily-IA phosphomonoesterase.* — Strongly supported by domain signatures, intact catalytic motifs (Asp11/Asp13/Thr15/Lys113/Asp168/Asp172), Mg^{2+} cofactor, and the two-domain C1-capped fold. 2. *Its default/primary predicted activity is 2-phosphoglycolate phosphatase (EC 3.1.3.18), acting in cytosolic glyoxylate/dicarboxylate metabolism as a 2-phosphoglycolate salvage enzyme.* — Supported by convergent KEGG/PANTHER/GO annotation, top homology to characterised *E. coli* Gph, and the intact downstream glycolate-utilisation pathway. 3. *The enzyme functions intracellularly (cytosol).* — Supported by GO and absence of signal/TM features.

Tempered / partially refuted 4. *PP_1907 is THE 2-phosphoglycolate phosphatase of *P. putida*.* — **Tempered.** PP_1907 is one of three paralogs; **PP_0416** (43% identity to Gph; explicitly named phosphoglycolate phosphatase) is a better candidate for the principal enzyme. PP_1907's exact substrate is unconfirmed and may be broader. 5. *STRING network proves PP_1907 acts in glycolate metabolism.* — **Refuted as specific evidence.** The network is a paralog/co-annotation artifact shared identically by all three K01091 genes.

6. Evidence Quality and Limitations

- **No experimental (ECO:0000269) data exist for PP_1907;** all annotations are imported (ECO:0000313) or automatic (ECO:0000256/ARBA). The functional picture is built entirely from **structure, evolution and bioinformatics**.
- The strongest *experimental anchor* is the characterisation of the orthologous **E. coli Gph** (Pellicer et al., 2003,

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transferred by homology.

- **Paralog ambiguity** (three K01091 genes) means enzyme-specific substrate and physiological roles cannot be assigned with certainty from sequence alone.
- The AlphaFold model for Q88LM3 could not be retrieved during this study, so active-site geometry was inferred from InterPro/Gene3D domain assignments and motif analysis rather than direct 3D inspection.

Future directions

- **Direct biochemistry:** recombinant expression of PP_1907 (and PP_0416, PP_0094) and phosphatase substrate profiling (e.g., against 2-phosphoglycolate, sugar phosphates, nucleotide monophosphates, phosphoglycerate) with kinetic constants — the definitive test of substrate specificity, following the HAD substrate-screening approach of Kuznetsova et al. (2006,

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- **Genetics:** single and combinatorial deletions of the three paralogs, testing bleomycin/oxidative-stress sensitivity and growth on glycolate, to resolve which paralog performs the physiological 2-phosphoglycolate salvage and whether they are redundant.
- **Structure:** experimental structure or validated AlphaFold model with docked 2-phosphoglycolate to examine cap-domain substrate-recognition residues.

7. Key References

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- Biswas T, et al. *The tail of KdsC: conformational changes control the activity of a HAD superfamily phosphatase.* 2009.

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- Dos Santos VAPM, Heim S, Moore ERB, Strätz M, Timmis KN. *Insights into the genomic basis of niche specificity of Pseudomonas putida KT2440.* Environ Microbiol. 2004.

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Database resources consulted: UniProt (Q88LM3), NCBI (AAN67525.1/WP_010952931.1), KEGG (ppu:PP_1907; K01091), InterPro/Pfam/Gene3D/PANTHER/SUPFAM, eggNOG (COG0546), STRING (160488.PP_1907), BioCyc.